AHMAD ISHAMUL AYADY AKMAL, S.T.

Process Modeling & Simulation | Metallurgical Data Analytics | Engineering Design
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PROFESSIONAL SUMMARY

Metallurgical engineer (M.Eng. candidate) with a multidisciplinary electrical–metallurgical background, specialising in **numerical process modeling (CFD/FEA with ANSYS Fluent and Fortran-based custom code), data-driven analysis of process and laboratory data (Python, Microsoft Fabric), and engineering design** of process equipment and systems. Seeking an office- or lab-based role in **process/metallurgical engineering, process simulation, data analytics, or engineering design** to support copper, gold and nickel downstream (RKEF / HPAL) operations.

CORE COMPETENCIES

Simulation & Modeling: Numerical modeling of multiphase flow, heat transfer and FEA (ANSYS Fluent / Workbench), custom code development in Fortran, continuous-casting & inclusion modeling, model validation against plant data

Data Analysis: Python (data processing, analysis, visualisation), machine learning & deep learning, Microsoft Fabric / Azure, Git/GitHub, process-parameter correlation, technical KPI tracking.

Engineering Design: Process-equipment & system design, technical documentation.

Multidisciplinary Edge: Electrical power systems, power electronics and control fundamentals applied to furnace power, energy efficiency and process instrumentation

Metallurgy & Lab: Nickel, steel & stainless-steel production routes; non-metallic inclusions; SEM-EDS, optical microscopy, metallographic preparation, QC documentation

EDUCATION

M.Eng. Metallurgical Engineering — University of Science and Technology Beijing (USTB) Oct 2024 – Jun 2027 (expected)
LPDP–Tsingshan Group Scholarship (Indonesia Endowment Fund for Education)

- **Thesis:** Numerical Modeling on the Evolution and Distribution of Inclusions in Continuous Casting of Heavy Rail Steel.
- **Coursework:** Advanced Metallurgy, Steelmaking Processes, Computational Fluid Dynamics.

B.Eng. Electrical Engineering — University of Jember Aug 2019 – Jan 2023
GPA 3.66 / 4.00 | Djarum Beasiswa Plus Awardee

- **Thesis:** Power-electronics design (flyback converter for LED lighting), published in IJPEDS (Scopus-indexed, 2023).

PROFESSIONAL & RESEARCH EXPERIENCE

Metallurgical Research Intern — Tsingtuo Group Research Institute (Tsingshan Group), China Oct 2025 & Mar 2026
Stainless Steel R&D Division

- Analysed non-metallic inclusions in stainless-steel samples with SEM-EDS, classifying morphology and chemistry per GB/T characterisation standards.
- Cross-referenced laboratory results with production parameters (alloying, refining, casting) to trace how upstream process variables drive inclusion formation.
- Documented the full stainless-steel route — raw material processing, alloying, refining, continuous casting — through structured sessions with R&D engineers.

Site Support Equipment Group Leader — PT Hasnur Riung Sinergi, Tapin, South Kalimantan Sep 2023 – Sep 2024
Fresh Graduate Development Program | Mining Equipment Maintenance

- Supervised a cross-functional team of 17 mechanics and welders maintaining heavy mining equipment under SHE-compliant procedures.
- Designed and enforced structured preventive-maintenance schedules, using equipment availability and downtime data to prioritise work and reduce unplanned breakdowns.
- Standardised fault-documentation templates and cross-shift escalation, building a cleaner failure record that shortened root-cause diagnosis and MTTR.

Engineering Intern — PT Lentera Bumi Nusantara, Tasikmalaya Jan 2022 – Feb 2022
Renewable Energy Equipment

- Assembled a vertical-axis wind turbine, maintained five TSD 500 turbines and studied their power-system controller.

Laboratory Assistant, Electrical Energy Conversion Lab — University of Jember Mar 2021 – Dec 2022

- Designed trainer modules, authored the practicum manual and coordinated practicum sessions for six classes.

SELECTED PROJECTS

CFD Simulation of Inclusion Behaviour in Continuous Casting — USTB Oct 2024 – Present

- Built a multiphase numerical model (ANSYS Fluent with Fortran-based custom code) coupling turbulent flow, thermal gradients and inclusion nucleation/transport; validating results against industrial casting data from partner plants.

Engineering Design Projects — University of Jember 2021 – 2023

- Designed and tested a flyback LED driver (published, patent filed) and a solar-powered fishing-lamp prototype; grant Kemendikbudristek PPMW and LP2M research funding.

PUBLICATIONS & INTELLECTUAL PROPERTY

- "Switch mode power supply with flyback LED driver topology in public lighting systems energy supply," International Journal of Power Electronics and Drive Systems (IJPEDS), 2023.
- Patent application: flyback switch-mode power supply for LED drivers. Co-author of a registered-copyright book on energy storage technology (Univ. of Jember, 2020).

CERTIFICATIONS & TRAINING

- **Dicoding Indonesia, Data & Programming:** Python Programming; Data Analysis; Data Visualization; Data Processing Fundamentals; Data Science Fundamentals; Git & GitHub
- **Dicoding Indonesia, Machine Learning & AI:** Machine Learning for Beginners; Applied Machine Learning; Building Machine Learning Systems; Deep Learning Fundamentals; AI Fundamentals; Generative AI Fundamentals
- **Dicoding Indonesia, Cloud & Applied Data Science:** Data Science with Microsoft Fabric; Building Generative AI Applications with Microsoft Azure; Cloud & Generative AI Fundamentals on AWS (2026)
- Data Science & Machine Learning Bootcamp — DQLab | Machine Learning & Deep Learning — Kredensial Mikro Mahasiswa Indonesia
- Machine Learning Path — Studi Independen (Kampus Merdeka), PT Presentologics
- BNSP Certification: PLC-Based Electronic Equipment Operation

SELECTED ACHIEVEMENTS & LEADERSHIP

- Top 5, Energy Transition Innovation Challenge — Nexus Energy & Ministry of Energy and Mineral Resources (2022); Top 5, Think Efficiency — Shell × SRE (2023)
- Rank 2, Outstanding Student (Mahasiswa Berprestasi), University of Jember (2022); East Java representative, Indonesia–Korea Youth Exchange Program, Kemenpora (2023)
- Chairman, UKM Einstein research club (2021): led 4 departments to 84% program completion and 26 national/international awards

LANGUAGES

Indonesian (native) | English — Duolingo English Test 110 (B2+) | Mandarin — HSK 3